

Comprehensive Exam (Closed Book)

PART-A

Time-: 14:00 – 17:00

Max Marks-: 25

Date: 12-05-2022

ID No.:

Name:

Instructions: - Questions 1 to 5 carry one mark each. For questions 6 to 10, each right answer carries 2 marks while every incorrect answer carries -1 mark. Read the questions carefully and write the answers in the space provided. Rough work must be carried out on the supplementary sheet provided separately. Both question paper as well as the answer sheet must be submitted for evaluation. Overwriting or ambiguity will not be considered for evaluation.

Answers

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15

- Which of the following option is correct for ideal op-amp?
 - Input impedance is zero, voltage gain is infinite and output impedance is zero.
 - Input impedance is infinite, voltage gain is infinite and output impedance is zero.
 - Input impedance is infinite, voltage gain is zero and output impedance is zero.
 - Input impedance is infinite, voltage gain is infinite and output impedance is infinite.
- Decreasing the gain of the following circuit shown in Figure 1 can be achieved by varying which parameters?
 - Reducing the amplitude of the input voltage.
 - Increasing the value of feedback resistance.
 - Increasing the value of input resistance
 - Removing feedback resistance
- For an op-amp having a slew rate of $SR = 2 \text{ V}/\mu\text{s}$, pick the correct option for the maximum closed-loop voltage gain that can be used when the input signal varies by 0.25 V in $10 \mu\text{s}$?
 - 20
 - 40
 - 60
 - 80
- Which of the following is not a part of a Phase locked loop (PLL)
 - Multiplier
 - Adder
 - Integrator
 - Oscillator
- Identify the the circuit functionality given in Figure 2. Assume the input is a sine wave.
 - Voltage divider
 - HW rectifier
 - FW rectifier
 - Square root circuit

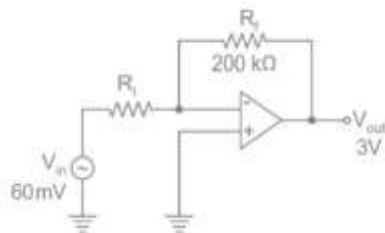


Figure 1: circuit for Qn 2

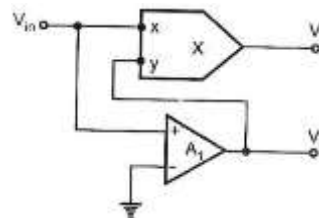


Figure 2: circuit for Qn 5

- What are the values of resistors (R) and feedback resistor (R_F), if resolution required is 0.5 V for 4-bit $R/2R$ ladder type DAC?
 - $8 \text{ k}\Omega$ and $10 \text{ k}\Omega$
 - $10 \text{ k}\Omega$ and $12 \text{ k}\Omega$
 - $12 \text{ k}\Omega$ and $10 \text{ k}\Omega$
 - $12 \text{ k}\Omega$ and $8 \text{ k}\Omega$
- For a particular flash type 8-bit ADC, the conversion time is $9 \mu\text{s}$. Find the maximum frequency of an input sine wave that can be digitized?
 - 17.6 kHz
 - 111.1 kHz
 - 434.02 Hz
 - 69.07 Hz
- An amplifier has an open loop gain of 100, an input impedance of $1 \text{ k}\Omega$ and an output impedance of 100Ω . If a feedback network with a feedback factor of 0.49 is connected in a voltage series feedback mode then new input and output impedances are
 - 100Ω & $1 \text{ k}\Omega$
 - 50Ω & $1 \text{ k}\Omega$
 - $50 \text{ k}\Omega$ & 2Ω
 - $100 \text{ k}\Omega$ & 1Ω

9. In the circuit shown in Figure 3, the R_1 , R_2 and R_3 values are $16\text{ k}\Omega$, $29\text{ k}\Omega$ and $43\text{ k}\Omega$, respectively. Pick the correct option for the input impedance of the circuit?
- (A) $-23.72\text{ k}\Omega$ (B) $-26.09\text{ k}\Omega$ (C) $-21.35\text{ k}\Omega$ (D) None of these
10. The triangular wave generator circuit is shown in Figure 4 whose frequency $f_0 = 5\text{ kHz}$ and $V_{\text{out(pp)}} = 10.5\text{ V}$. The op-amp saturation voltage is $\pm 15\text{ V}$. Assume $R_2 = 10\text{ k}\Omega$ and $C_1 = 0.01\text{ }\mu\text{F}$. Find the resistance values of R_1 and R_3 .
- (A) $40\text{ k}\Omega$ and $20\text{ k}\Omega$ (B) $20\text{ k}\Omega$ and $40\text{ k}\Omega$ (C) $10\text{ k}\Omega$ and $20\text{ k}\Omega$ (D) $20\text{ k}\Omega$ and $10\text{ k}\Omega$
11. For the comparator circuit shown in Figure 5, forward diode voltage drop is $V_\gamma = 0.7\text{ V}$ and reverse Zener diode voltage is 5.1 V . If hysteresis width is 1 V , then value of R_2 is
- (A) $10.6\text{ k}\Omega$ (B) $11.6\text{ k}\Omega$ (C) $23\text{ k}\Omega$ (D) $24\text{ k}\Omega$
12. The scale factor of multiplier IC AD633 is 10. In the circuit shown in Figure 6, if $R_2 = 4R_1 = 4\text{ k}\Omega$, the scale factor of IC AD633 has been changed to
- (A) 2 (B) 5 (C) 10 (D) 20
13. What is the voltage gain of the circuit shown in Figure 7.
- (A) $(R_1R_3) / (R_2R_G)$ (B) $(R_2R_3) / (R_1R_G)$ (C) $(R_2R_G) / (R_1R_3)$ (D) $(R_1R_G) / (R_2R_3)$
14. Inductance can be simulated using the circuit shown in Figure 8. For $C = 0.1\text{ }\mu\text{F}$, $R_1 = 10\text{ k}\Omega$, $R_2 = 10\text{ }\Omega$, at which frequency, maximum Q is obtained, and what is the value of Q_{max} .
- (A) 10 kHz & 50 (B) 160 Hz & 50 (C) 80 Hz & 25 (D) 40 Hz & 25
15. For the circuit shown in Figure 9, $R_4 = R_5 = R_6$. The slope of the voltage transfer characteristics for $V_i > 0$ and $V_i < 0$ respectively are
- (A) 1 & -1 (B) 0 & 1 (C) 0 & -1 (D) -1 & 1

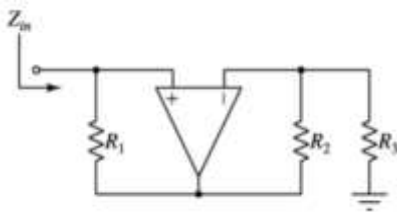


Figure 3: circuit for Qn 9

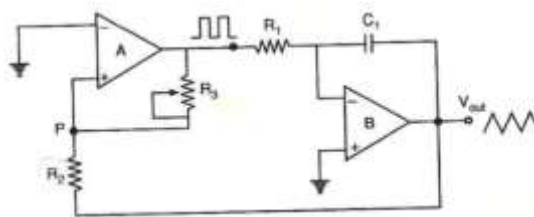


Figure 4: circuit for Qn 10

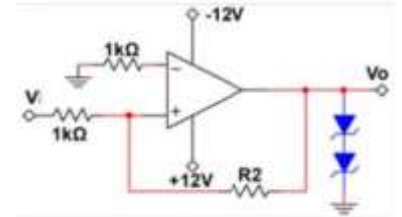


Figure 5: circuit for Qn 11

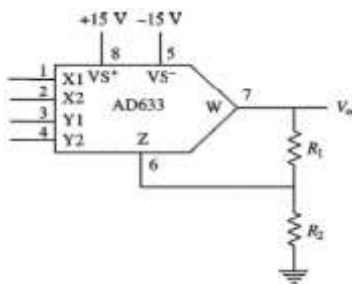


Figure 6: circuit for Qn 12

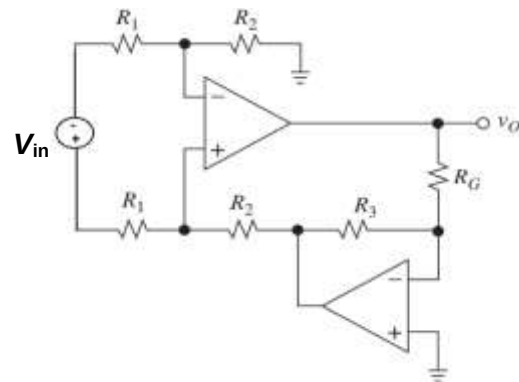


Figure 7: circuit for Qn 13

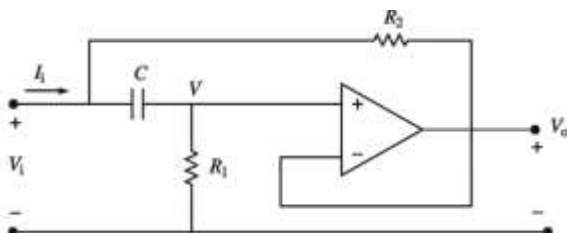


Figure 8: circuit for Qn 14

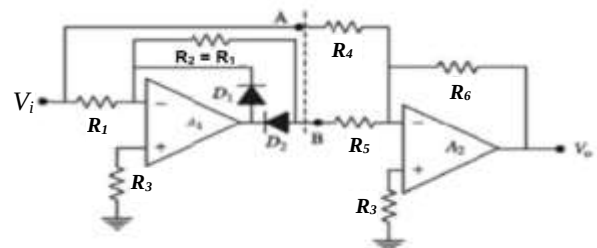


Figure 9: circuit for Qn 15